

Watching Your AI in Production

Three questions, three tools · CloudWatch Logs and Alarms · CloudTrail · X-Ray · Lake Formation

Three questions, three tools

The question	The tool
What is happening? — errors, latency, token use	CloudWatch
Who did what, and when? — every API call	CloudTrail
Where is it slow? — one request across services	X-Ray



They're easy to mix up. Start from the question you need answered, not the tool.

CloudWatch Logs

Logs record what happened, line by line.

-  Lambda writes its logs here automatically
-  Bedrock model invocation logging can record every prompt and response
-  Logs Insights lets you search them with a short query



"Show me the 20 slowest model calls today."

```
fields @timestamp, modelId, @duration
| sort @duration desc
| limit 20
```

CloudWatch Alarms and RUM

An alarm watches a number and tells you when it crosses a line.

Watch this metric	Alarm when...	Then
InvocationThrottles	Above 10 in 5 minutes	Email the team (SNS)
InvocationLatency	Average above 5 seconds	Email the team
OutputTokenCount	Sudden jump	Check for runaway costs



CloudWatch RUM measures what real users experience in their browser — page load times and errors.

CloudTrail: who did what

CloudTrail records every API call — who, what, when and from where.



"Who deleted the guardrail on Tuesday?" → CloudTrail answers that in seconds.

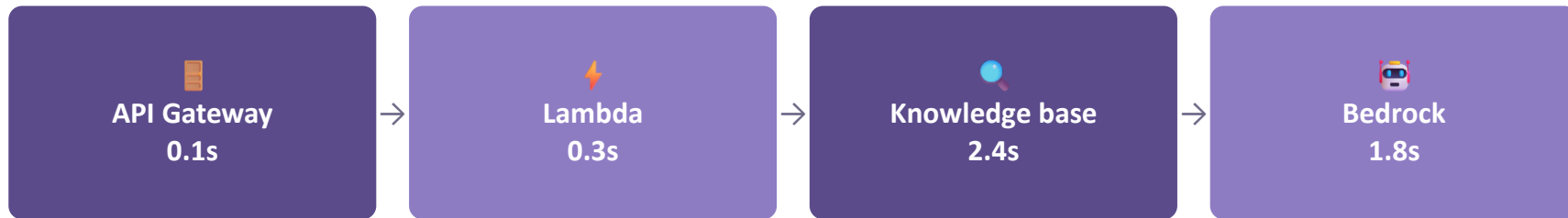
You want to know...	Use
Who called the model, and when	CloudTrail
What was asked, and what it replied	Bedrock model invocation logging



CloudTrail shows that a model was called — not the prompt or the answer.

X-Ray: where is it slow?

X-Ray follows one request through every service it touches.



Users say the app is slow. Everyone blames the model.

X-Ray shows the knowledge base search takes the longest.



Measure before you optimise — the slow part is often not the one you'd guess.

Lake Formation: who sees which data

Controls access to data in your data lake — down to the column and row.

Person	Can see
Data scientist	Purchase history — without names or emails
Regional manager	Only rows for their own region
Fine-tuning job	Only the approved columns



For AI, this decides what data a model is ever allowed to learn from.

Key takeaways

- 1 CloudWatch: what's happening · CloudTrail: who · X-Ray: where
- 2 Turn on Bedrock invocation logging to see prompts and answers
- 3 Alarm on throttles, latency and tokens
- 4 Measure before you optimise
- 5 Lake Formation decides what data a model can learn from

Next · Part 3: securing your AI on AWS